EcoDashboard

Business Requirements Document

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1. Business Problem

Individuals who want to stay informed about environmental conditions currently have to check multiple disconnected sources - one site for air quality, another for climate news, another for conservation status of endangered species. There is no single, free, self-hostable dashboard that aggregates this information into one live view.

2. Business Objectives

- ❖ Aggregate real-time air quality data (AQI, PM2.5) into a single dashboard view.
- ❖ Surface current environmental and climate news without requiring the user to visit multiple news sites.
- ❖ Raise awareness of critically endangered species by displaying population, trend, and conservation status.
- ❖ Keep the data current automatically, without requiring manual refreshing of multiple sources.
- ❖ Provide a free, self-hostable alternative to fragmented environmental-awareness tools.

3. Stakeholders

Stakeholder	Interest / Role
End User (Primary)	Wants a quick, current snapshot of environmental conditions and conservation news without visiting multiple sites.
Developer / Maintainer (Jordan)	Builds and maintains the system; wants a maintainable, portfolio-quality full-stack architecture.
External Data Providers	Open-Meteo (air quality) and RSS news sources whose data availability directly affects dashboard reliability.

4. Current State vs. Future State

Step	Current State (Manual)	Future State (EcoDashboard)
Air quality check	User visits a separate air-quality site to check AQI/PM2.5.	Dashboard pulls live AQI and PM2.5 from the Open-Meteo Air Quality API automatically.
Climate news	User browses multiple news sites for environmental stories.	Custom Express API aggregates RSS feeds into a single news panel, with fallback content if a source is down.
Species awareness	User researches conservation status separately, if at all.	Dashboard displays tracked species (e.g. Vaquita, Amur Leopard) with population, trend, and status indicators.

Keeping data current	User manually re-checks each source.	Dashboard auto-refreshes every 5 minutes, with a manual refresh option available.
Deployment	N/A -- no existing tool.	Containerized via Docker Compose with Nginx serving the front end.

5. Functional Requirements

ID	Requirement	Priority
FR-1	System shall retrieve live AQI and PM2.5 data from the Open-Meteo Air Quality API.	Must have
FR-2	System shall display an air-quality status indicator derived from current readings.	Must have
FR-3	System shall auto-refresh air quality data on a fixed interval (5 minutes).	Must have
FR-4	System shall aggregate environmental/climate news via a custom Express API using RSS parsing.	Must have
FR-5	System shall display fallback news content when RSS/API sources are unavailable.	Should have
FR-6	System shall display name, estimated population, population trend, and status for tracked endangered species.	Must have
FR-7	System shall display a global CO2 emissions indicator on the dashboard.	Should have
FR-8	System shall support a manual refresh action in addition to auto-refresh.	Should have
FR-9	System shall be deployable as a containerized stack (Docker Compose + Nginx).	Should have

6. Non-Functional Requirements

- Reliability: news panel must degrade gracefully (fallback content) rather than break when an RSS source is unreachable.
- Performance: air quality data should refresh without a full page reload or blocking the UI.
- Portability: application must run identically via `npm run dev` locally and via Docker Compose, with Nginx handling static serving/reverse proxy in the containerized version.
- Usability: dashboard must present all three data categories (air quality, news, species) without requiring navigation to separate pages.

7. Assumptions and Constraints

- Application relies on third-party data availability (Open-Meteo, RSS sources); it does not persist historical data in a database.
- Endangered species data is currently a fixed/reference set (e.g. Sumatran Orangutan, Javan Rhino, Vaquita, Amur Leopard), not a live-updating feed.
- Solo developer project -- no cross-team dependencies.

❖ QA automation (Selenium, REST Assured, Jest) and cloud infrastructure (Terraform/AWS) work exist as separate companion repositories, not inside the core application repo.

8. Success Criteria

❖ A user can view current AQI, PM2.5, and an air-quality status indicator within 2 seconds of dashboard load.

❖ News panel displays current articles or valid fallback content in 100% of test cases, including simulated RSS outage.

❖ Full stack starts successfully via a single `docker-compose up --build` command.

9. User Story Backlog

As a...	I want to...	So that...	Priority
User	see current AQI and PM2.5 on load	I know today's air quality without visiting another site	High
User	see a status indicator, not just raw numbers	I can interpret air quality at a glance	High
User	see recent environmental news in one place	I don't have to browse multiple news sites	High
User	still see relevant content if the news source is down	the dashboard never looks broken or empty	Medium
User	see which species are critically endangered and their trend	I understand conservation status at a glance	Medium
User	manually refresh the dashboard	I can get the latest data on demand, not just every 5 minutes	Low

10. Out of Scope

❖ Historical data storage / database persistence.

❖ User accounts, authentication, or personalized dashboards.

❖ Live-updating (API-driven) endangered species data -- currently a fixed reference set.